EWB CLOUD LABS / AWS NETWORKING

AWS VPC Peering Implementation & Verification Report

Build • Test • Explain

DETAIL	INFORMATION
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Batch	Batch 1
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Programme	B.Tech — Computer Science & Engineering (IoT)
AWS Region	Asia Pacific (Mumbai) — ap-south-1
Project	Private connectivity between two VPCs using VPC peering
Evidence date	31 October 2026

Outcome. Two non-overlapping VPCs were connected using an Active VPC peering connection. App-A and App-B exchanged private ICMP traffic in both directions with 0% packet loss. App-B was accessed through the public bastion using SSH. A controlled ICMP security-group failure was demonstrated, followed by successful recovery.

Prepared against the EWB CLOUD LABS Student Project Workbook — AWS VPC Peering, Project 01 / Oct 2026.

01 | Executive Summary

This project implements a private network connection between two Amazon Virtual Private Clouds (VPCs) in the same AWS account and Mumbai Region. VPC A contains a public bastion and private App-A. VPC B contains private App-B and has no Internet Gateway. Non-overlapping IPv4 CIDR blocks and a VPC peering connection provide private inter-VPC routing.

Route tables provide forward and return paths, while security groups restrict inbound SSH and ICMP to the required source addresses. The supplied AWS console evidence shows all three EC2 instances Running with 3/3 status checks passed. Terminal evidence shows SSH access to App-B through the bastion, successful four-packet ping tests in both directions, and controlled failure followed by recovery.

Evidence limitations: The initial baseline ping failure screenshot was not supplied. Cleanup has not yet been performed; the EC2 overview shows the instances Running. Neither item is represented as completed in this report.

Objectives

- Design two non-overlapping VPC CIDR ranges.
- Keep both application instances private with no public IPv4 addresses.
- Provide administration through a public bastion.
- Activate VPC peering and configure forward and return routes.
- Verify bidirectional private connectivity and test the impact of a security-group rule change.

02 | Architecture and Address Plan

The student laptop connects to the bastion through its public IPv4 address. The bastion provides SSH access to private instances. App-A and App-B communicate directly using private IPv4 addresses through VPC peering.

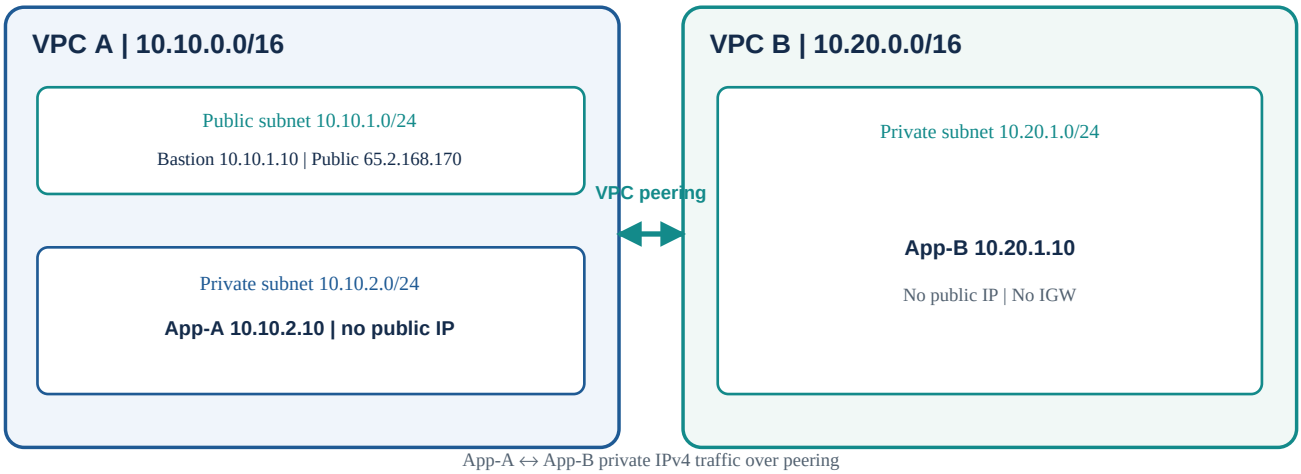


Figure 1. Logical architecture based on the workbook and recorded resource configuration.

RESOURCE	CIDR / ADDRESS	PURPOSE
VPC A	10.10.0.0/16	Bastion and App-A
VPC B	10.20.0.0/16	App-B; no Internet Gateway
ewb-a-public	10.10.1.0/24	Bastion subnet
ewb-a-private	10.10.2.0/24	App-A subnet
ewb-b-private	10.20.1.0/24	App-B subnet

The VPC ranges 10.10.0.0/16 and 10.20.0.0/16 do not overlap. Each /24 subnet is contained within its parent VPC.

03 | AWS Resource Inventory

Resource IDs are transcribed from the AWS console and implementation record. Region: ap-south-1 (Mumbai).

RESOURCE	NAME / ID	CONFIGURATION
VPC A	vpc-0f4d2bbca7c16900e	10.10.0.0/16
VPC B	vpc-0c4f04cf386b90f0b	10.20.0.0/16
Public subnet A	subnet-0aa3e91aaa0d0d183	10.10.1.0/24
Private subnet A	subnet-071b9ed6536cd7384	10.10.2.0/24
Private subnet B	subnet-0d5a6c3decaabf4aa	10.20.1.0/24
Internet Gateway	igw-0a2fadbe50ab77b0e	ewb-igw-a; attached to VPC A
Peering	pcx-0cb301468ddaa51b2	ewb-a-b-peer; Active
Public RT	rtb-09fcc0f404df00f25	ewb-rt-a-public
VPC A private RT	rtb-073d4eab1847dcfff	Main route table; associated with private subnet A
VPC B private RT	rtb-016b185358f36231b	ewb-rt-b-private
Bastion SG	sg-08926e9f3c3493629	ewb-sg-bastion
App-A SG	sg-0f7cb094c343d5a43	ewb-sg-app-a
App-B SG	sg-04fcafdbc005a6521	ewb-sg-app-b

INSTANCE	INSTANCE ID	PRIVATE / PUBLIC ADDRESS
ewb-bastion	i-09ad478e29fe394c6	10.10.1.10 / 65.2.168.170
ewb-app-a	i-0418fd8ff5009fdee	10.10.2.10 / none
ewb-app-b	i-016b7d1459239cc06	10.20.1.10 / none

All instances use Amazon Linux 2023, t3.micro and key pair ewb-lab. The EC2 screenshot shows Running and 3/3 status checks passed. The private key is not included in this report.

04 | Route Tables and Peering

The Active peering connection is the network link; routes must be configured on both VPC sides. The route-table screenshot identifies the route table IDs and subnet associations. The recorded route configuration is summarized below.

ROUTE TABLE	SUBNET ASSOCIATION	ROUTES
ewb-rt-a-public rtb-09fcc0f404df00f25	ewb-a-public 10.10.1.0/24	10.10.0.0/16 → local 0.0.0.0/0 → IGW 10.20.0.0/16 → pcx-0cb301468ddaa51b2
VPC A private main RT rtb-073d4eab1847dcfff	ewb-a-private 10.10.2.0/24	10.10.0.0/16 → local 10.20.0.0/16 → pcx-0cb301468ddaa51b2
ewb-rt-b-private rtb-016b185358f36231b	ewb-b-private 10.20.1.0/24	10.20.0.0/16 → local 10.10.0.0/16 → pcx-0cb301468ddaa51b2

Naming note: The AWS console shows rtb-073d4eab1847dcfff as the VPC A main route table and associated with the private subnet. Its display name is blank; this report uses the actual ID rather than claiming the console shows a custom name.

Traffic path explanation

Forward path: App-A sends traffic to 10.20.1.10. The VPC A private route table forwards 10.20.0.0/16 to the peering connection. App-B's security group permits ICMP Echo Request from 10.10.2.10/32.

Return path: VPC B's route table forwards 10.10.0.0/16 to the same peering connection, allowing the reply to return to App-A.

Bastion to App-B: The public subnet route table has 10.20.0.0/16 → peering, allowing the bastion's proxied SSH connection to reach App-B over its private address.

05 | Security Groups and SSH

SECURITY GROUP	INBOUND RULE	SOURCE
ewb-sg-bastion	SSH / TCP 22	Current student public IPv4 /32 (saved rule recorded as 49.37.45.10/32)
ewb-sg-app-a	SSH / TCP 22	10.10.1.10/32 (bastion)
ewb-sg-app-a	ICMPv4 Echo Request	10.20.1.10/32 (App-B)
ewb-sg-app-b	SSH / TCP 22	10.10.1.10/32 (bastion)
ewb-sg-app-b	ICMPv4 Echo Request	10.10.2.10/32 (App-A)

Outbound rules were left at the default allow-all setting for the lab. SSH and ICMP were not opened to 0.0.0.0/0. Both application instances have no public IPv4 address.

The same local ewb-lab.pem key was used. From Windows PowerShell, SSH to App-B succeeded through the bastion public address 65.2.168.170 and then to private address 10.20.1.10. The private key remained on the student's laptop and was not copied to an EC2 instance.

06 | Connectivity Verification

TEST	SOURCE → DESTINATION	OBSERVED RESULT	STATUS
Forward ping	App-A 10.10.2.10 → App-B 10.20.1.10	4 transmitted, 4 received; 0% loss	PASS
Reverse ping	App-B 10.20.1.10 → App-A 10.10.2.10	4 transmitted, 4 received; 0% loss	PASS
Private SSH	Laptop → Bastion → App-B	Shell prompt on 10.20.1.10	PASS
EC2 health	All three instances	Running; 3/3 checks passed	PASS

These results demonstrate bidirectional private connectivity and successful administrative SSH access through the bastion. Active peering alone would not prove reachability; the end-to-end terminal tests provide that evidence.

Baseline ping before peering

The workbook requires a baseline ping from App-A to 10.20.1.10 before peering and peer routes, expected to receive no replies. The student confirmed that no baseline failure screenshot is available. This baseline is therefore marked **not evidenced** and is not claimed as completed.

07 | Controlled Failure and Recovery

The App-B inbound ICMP Echo Request rule with source 10.10.2.10/32 was temporarily removed while the SSH rule was retained. A repeated ping from App-A to App-B then received no replies: 4 packets transmitted, 0 received, 100% packet loss. After restoring the exact ICMP rule, connectivity resumed.

Controlled failure — terminal evidence

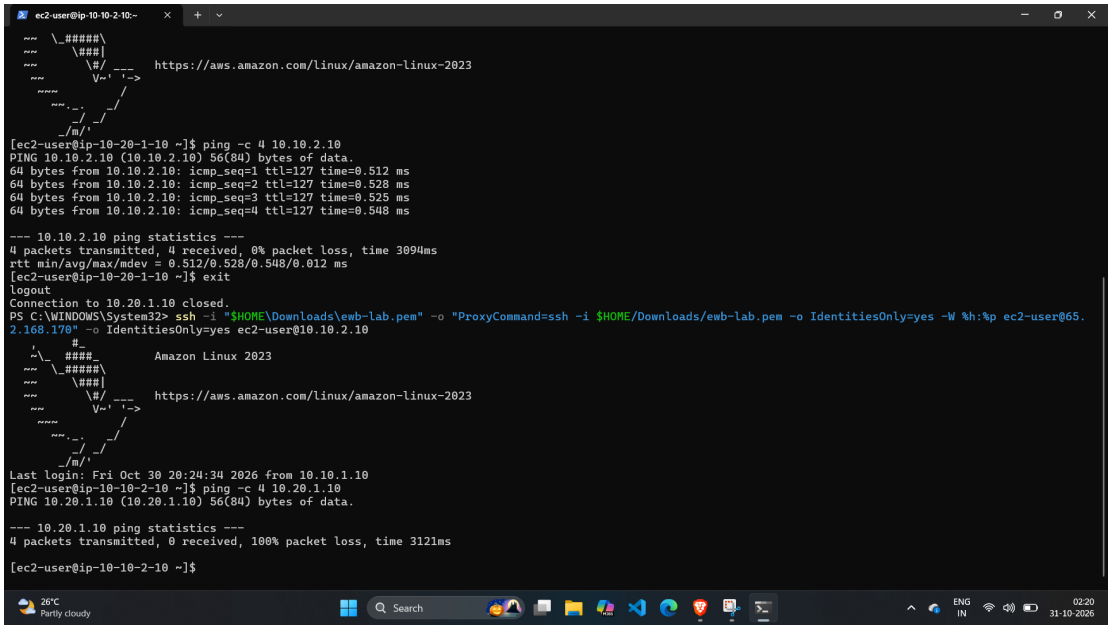


Figure 2. App-A ping to App-B fails while the App-B ICMP rule is removed.

Connectivity restored — terminal evidence

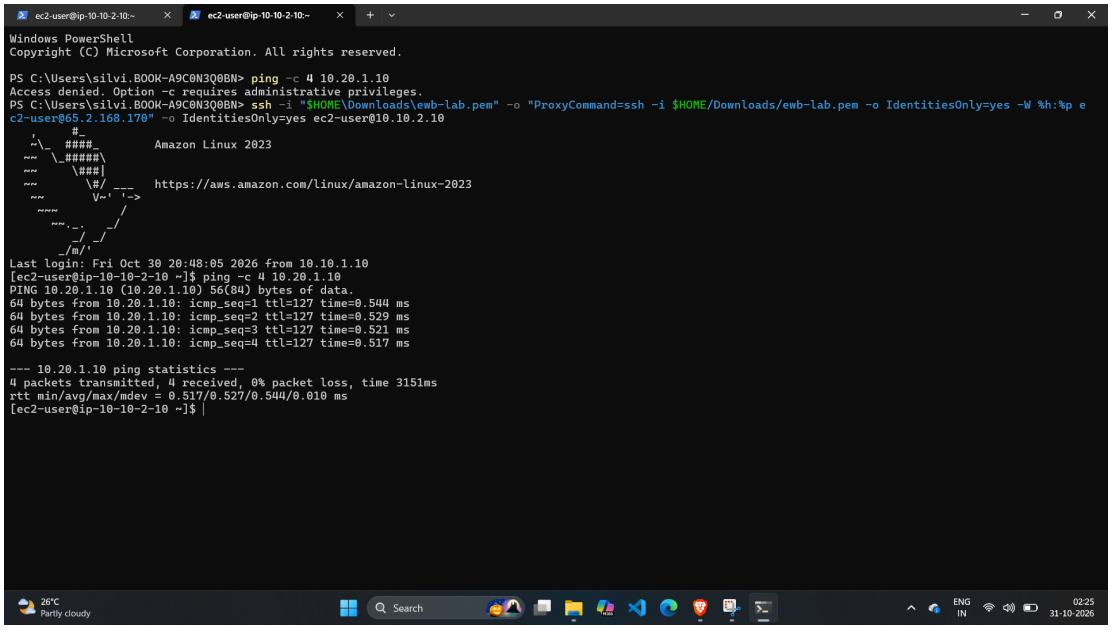


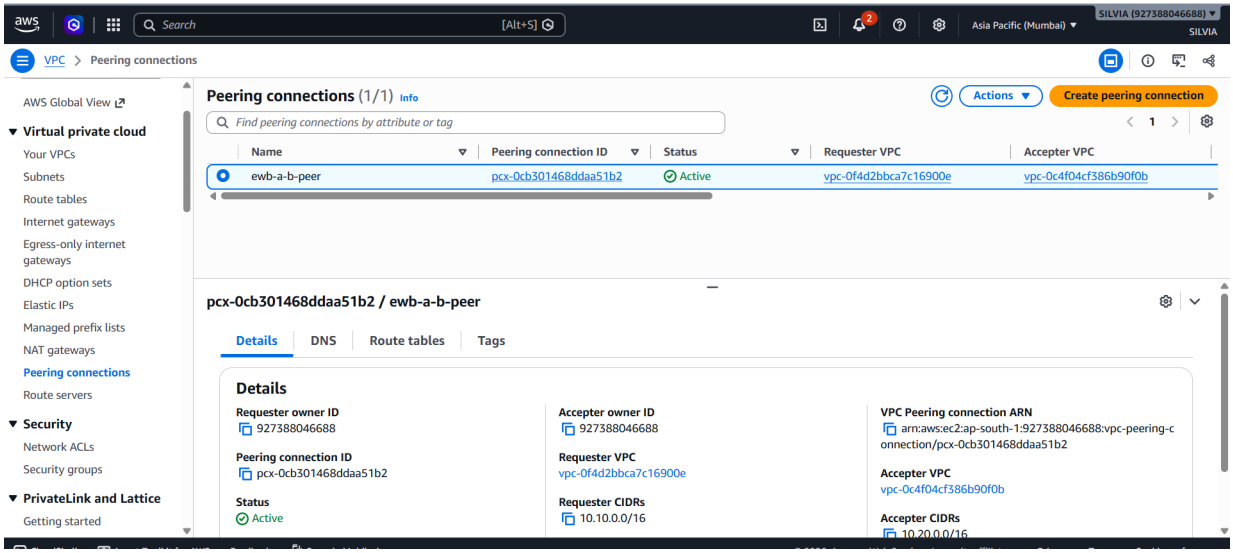
Figure 3. App-A ping succeeds again after the App-B ICMP rule is restored.

This test shows that a peering connection and routes are not sufficient by themselves: the destination security group must permit the required inbound traffic.

08 | Evidence Appendix

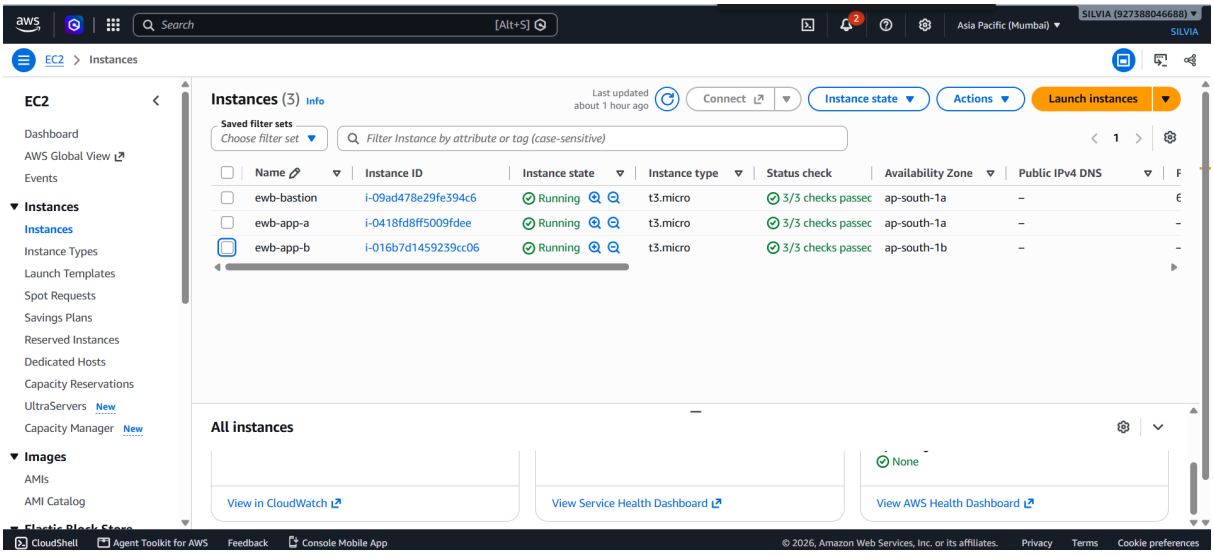
The screenshots below were supplied by the student. Each caption identifies the AWS console view or terminal test represented.

Active VPC peering connection



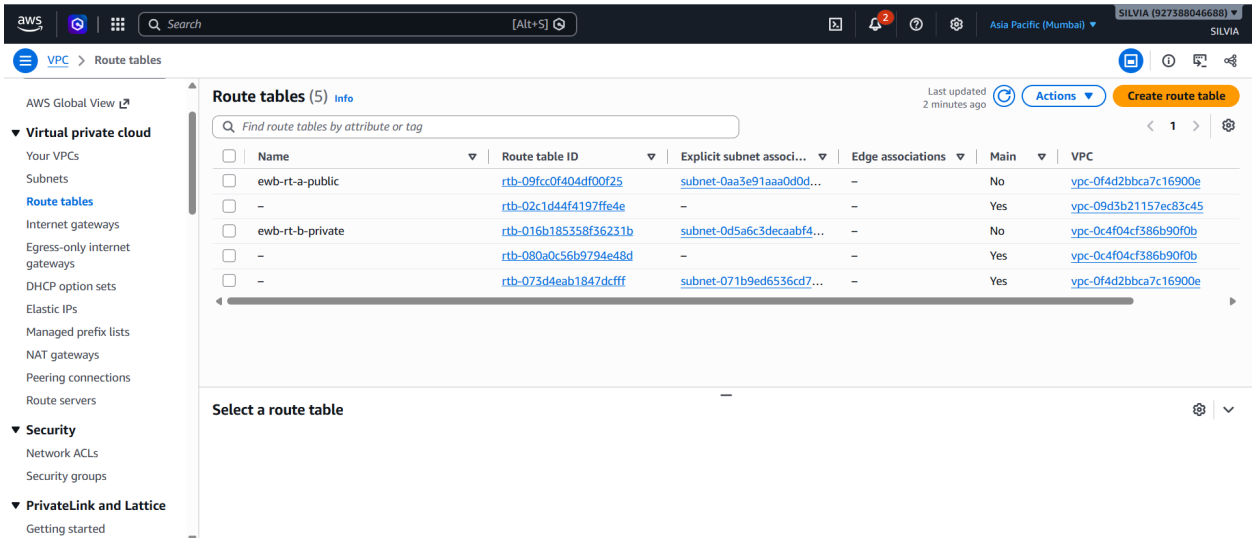
ewb-a-b-peer is shown Active, with VPC A as requester and VPC B as accepter.

EC2 instances overview



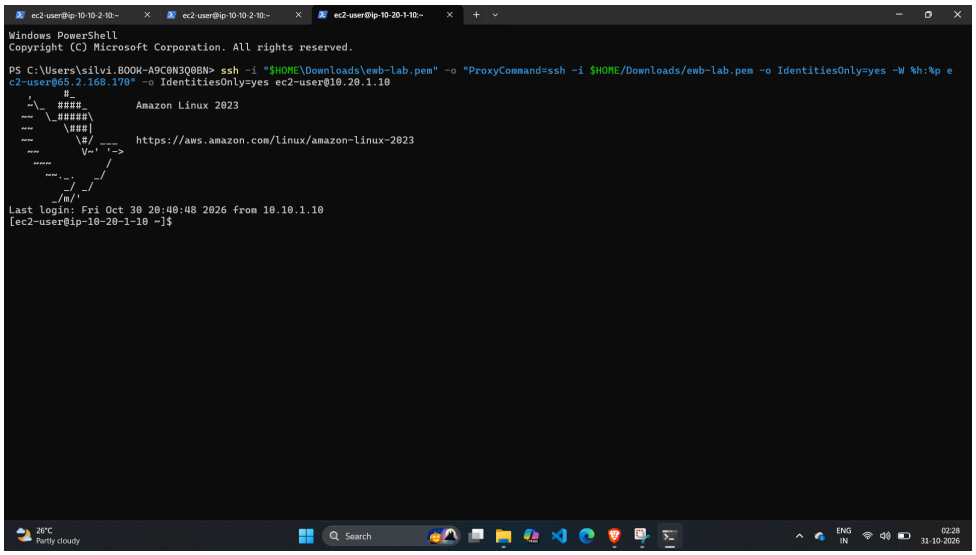
All three instances show Running and 3/3 status checks passed; private instances have no public IPv4 displayed.

Route table inventory



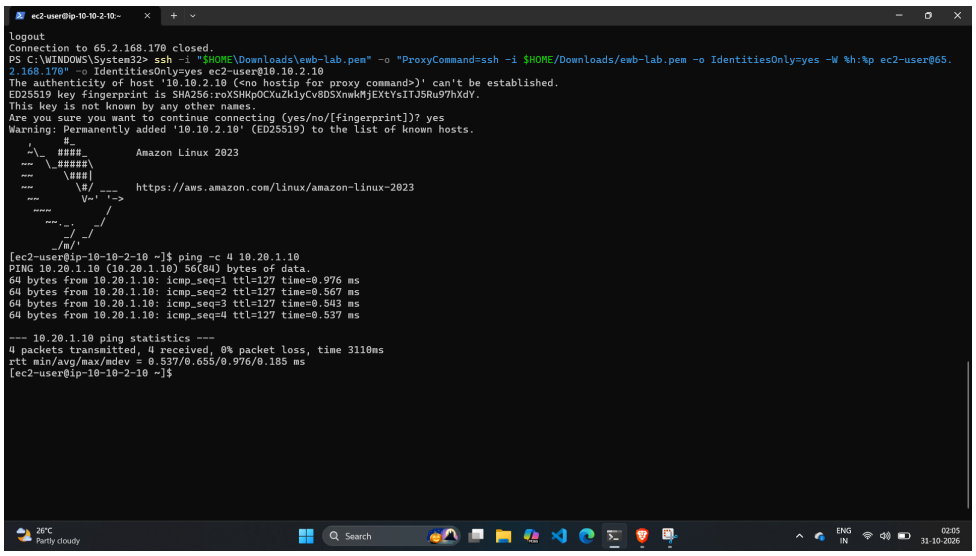
Console route table list and subnet associations. Route targets are detailed in Section 04.

SSH to App-B through bastion



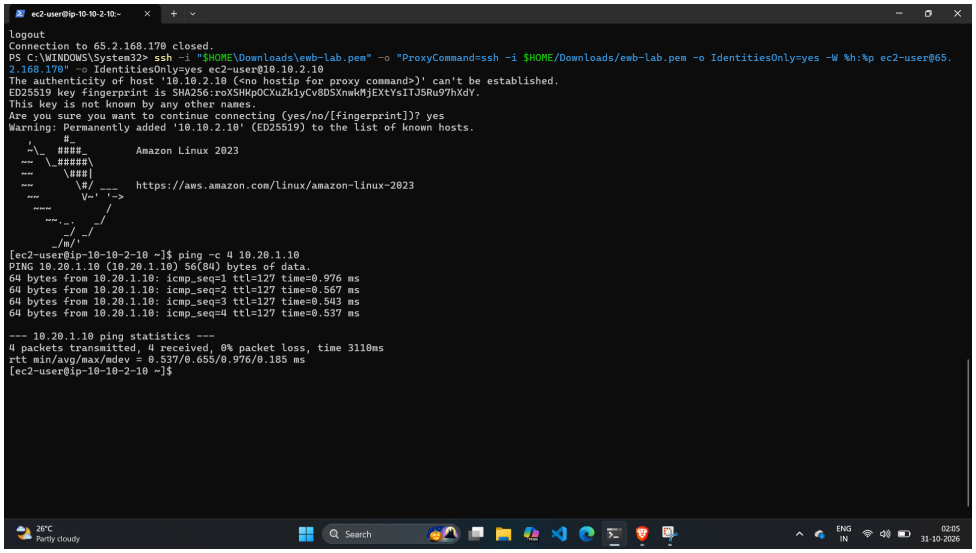
PowerShell command uses local key and bastion ProxyCommand; shell prompt identifies App-B.

App-A to App-B ping



Four replies from 10.20.1.10; 0% packet loss.

App-B to App-A ping



Four replies from 10.10.2.10; 0% packet loss.

09 | Learning Outcomes, Cleanup and Submission Status

Learning outcomes

• VPC peering provides private network connectivity, but route tables must provide both forward and return paths.
• The bastion subnet needs a peering route to reach App-B through proxied SSH.
• Security groups restrict access by protocol, port and source IP; /32 rules limit access to one host.
• An Active peering status does not alone prove application reachability; end-to-end tests are necessary.
• Controlled failure and recovery help isolate security-group behavior.

Cleanup status — pending

At the time of the latest confirmed status, all three EC2 instances were still Running. Cleanup has not been confirmed and must not be reported as complete. After preserving this report and all evidence, terminate the three lab instances; verify their lab EBS volumes and snapshots; release any lab Elastic IPs; delete the peering connection, security groups, subnets, route tables and Internet Gateway; then delete both project VPCs. Finally review EC2, VPC and Billing in ap-south-1 for remaining chargeable resources. AWS usage reporting may lag.

Final submission checklist

■ Add the baseline ping failure screenshot if it can be recovered; otherwise retain the explicit not-evidenced note.
■ Add inbound-rule screenshots for all three security groups if available.
■ Preserve the report and ZIP before cleanup.
■ Complete cleanup and capture proof; update this report if the workbook requires cleanup evidence.
■ Redact AWS account ID before making screenshots public on GitHub.
■ Upload the final PDF and repository link through the EWB Google Form.

Conclusion

The project successfully demonstrates private communication between VPC A and VPC B through VPC peering. Both directions of ICMP connectivity were verified with zero packet loss, App-B was accessed through the bastion using SSH, and a controlled security-group failure was followed by recovery. Baseline failure evidence and cleanup proof remain outstanding and are clearly identified for completion.

Source basis: EWB CLOUD LABS Student Project Workbook — AWS VPC Peering, Project 01 / Oct 2026; student-supplied AWS console and terminal screenshots; recorded resource configuration.